

I Feel Your Pain: a Haptic Interface for Improving Pain Literacy

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Abstract

There is no sensation more universal and misunderstood than pain. While pain presents itself in nearly every eukaryotic organism, it remains one of the most elusive disease states to express, let alone treat. Here, we introduce Pain by Numbers, a haptic, immersive storytelling interface that facilitates user recognition and communication of low-to-medium-intensity pain, in order to improve pain literacy for patients, physicians, and society-at-large.

CCS Concepts

• **Human-centered computing** → **Interface design prototyping**; • **Applied computing** → *Interactive learning environments*; *Life and medical sciences*.

Keywords

haptics, pain, expression, care, education, empathy

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1 Introduction

“The most intense feeling we know of, intense to the point of blotting out all other experiences, namely, the experience of great bodily pain, is at the same time the most private and least communicable of all.”

—Hannah Arendt, *The Human Condition* [1]

Humans possess a unique ability to experience not just their own pain but also the pain of others. Neuroimaging studies show that observing painful stimuli—whether through photographs of painful experiences or knowledge of a loved one’s pain—activates similar affective neural circuits as firsthand pain experience [6, 10]. Despite our ability to recognize when an experience may be painful, our pain empathy is limited by the subjective and inexpressible quality of painful experiences, creating a persistent gap between internal sensation and external expression. This challenge is evident

in clinical settings, where assessments of pain diverge significantly across observers. For instance, a study of pediatric patients found that children, parents, and nurses evaluated the same painful procedure differently—with nurses focusing on visible distress, parents relying on subjective interpretations, and children’s self-reports varying with age and communicative ability [8]. Standardized tools like the McGill Pain Questionnaire [7] attempt to mitigate these discrepancies by providing structured descriptors for pain’s sensory and emotional dimensions. However, such measures remain limited by language barriers, cultural influences, and individual variations in self-awareness and expression. Ultimately, this epistemic challenge—translating private experience into mutual understanding—continues to complicate both medical diagnosis and human empathy.

Beyond the challenges of communicating pain to others, an individual’s interpretation of pain profoundly shapes their subjective experience of pain. Pain Neuroscience Education (PNE) demonstrates this principle effectively. By teaching patients to reconceptualize pain as a protective output of the nervous system—rather than a reflection of tissue damage—PNE helps modify maladaptive cognitive patterns such as fear-avoidance behaviors and pain catastrophizing [9]. These cognitive shifts are clinically significant, as such beliefs are known contributors to the maintenance of chronic pain states and associated disability. Through its biopsychosocial approach, PNE enhances patients’ awareness of how their thoughts, emotions, and behaviors interact to modulate pain perception, ultimately empowering them to engage more adaptively with their condition.

The social dimension further modulates both the subjective experience of pain and its underlying neural processing [2, 3]. Empirical studies demonstrate that even simple acts—such as viewing images of loved ones—can activate reward-related brain regions and attenuate pain responses [11]. This phenomenon aligns with the adage that “pain shared is pain divided,” highlighting how social support can transform suffering through biopsychosocial mechanisms.

Integrating biopsychosocial research on experience and expression of pain, we developed Pain by Numbers: an immersive storytelling platform that uses a haptic interface consisting of Datafeel dots (generating vibration, heat, cold, and light) to simulate low-to-medium intensity pain experiences. By leveraging shared somatosensory pathways between pain and touch, our system enables users to “feel” canonical pain sensations and then select or describe the one that most closely matches their own. Designed for use in early-intervention care and preventative care settings, Pain by Numbers aims to improve pain literacy and empathy. Unlike traditional pain scales, our novel system circumvents linguistic barriers by grounding pain literacy in embodied simulation—making the private experience of pain tangible, communicable, and actionable.

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2 Method

Pain by Numbers is an immersive storytelling platform and haptic interface that explores how we experience, interpret, and communicate pain. Inspired by Hannah Arendt’s assertion that storytelling allows private experience to become publicly legible, we guide our user through three kinds of narrative experiences: (1) descriptive pain, which introduces users to the vocabulary of pain (e.g., “stabbing,” “flashing,” “pulling”); (2) contextualized pain, which reveals how self-beliefs and societal narratives about pain—such as the systemic dismissal of Black women’s pain—affect one’s experience of pain; and (3) chronic pain, which centers the slow, cumulative buildup of discomfort that may first escape acute recognition, e.g. in conditions such as fibromyalgia.

3 Prototype

We designed a haptic sleeve capable of producing sensations simulating low-to-medium intensity pain [5]. The sleeves hold four DataFeel dots within a 3D-printed enclosure with hardware that controls the activation and setting of the dots. The armband consists of a Lycra performance fabric and thermal insulation fabric to avoid unintended heat diffusion. The armbands include elastic Velcro for consistent skin contact across forearms of varying sizes. We selected the forearm for prototyping and evaluation due to its balance of thermal sensitivity and ease of access.

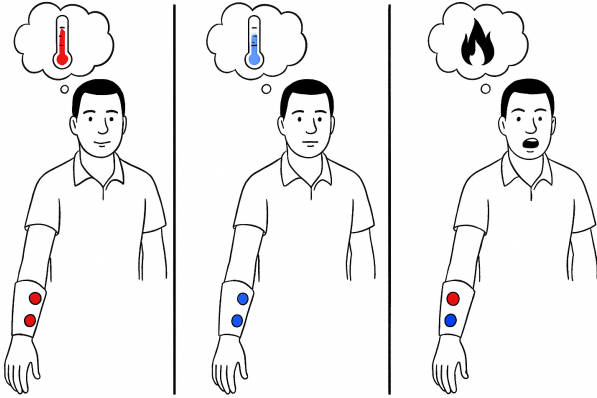


Figure 1: Simplified sketch of prototype armband, where red can reach up to 45 degrees Celsius and blue can lower down to 15 degrees Celsius.

3.0.1 The Thermal Grill Illusion (TGI). We simulated the sensation of pain using the thermal grill illusion (TGI). The thermal grill illusion is a sensory phenomenon in which interlaced warm and cool stimuli (typically 40°C and 20°C respectively), when applied simultaneously to the skin, produce a paradoxical sensation of burning heat without inflicting harm to the skin [4]. The illusion arises because the brain processes the conflicting thermal inputs in a nonlinear way: it suppresses the perception of cold, while enhancing the perception of heat and activating pain pathways, particularly those associated with burning sensations.

Pain Type	DataFeel Encoding
Burning	Sustained stimuli
Stabbing	Intense, medium-pulse stimuli
Dull	Low pulsed stimuli
Pins and needles	Quick stimuli, alternating locations
Numbness	Cold, sustained stimuli

Table 1: Example mappings of pain types to multisensory DataFeel encodings

3.0.2 Narrative Experience. The Pain-by-Numbers platform features a mosaic that slowly reveals itself as an artistic rendering of a specific pain experience as users experience that pain from multiple narrative perspectives. After being presented with a blank canvas, the user will start to experience a sensation of pain, revealing a portion of the mosaic as they listen to an audio narrative.

These audio narratives are recollections of real-life experiences of pain. We used the DataFeel toolkit to simulate low-to-medium intensity sensations like burning, throbbing, or stabbing through combinations of vibration, heat, cold, and light at appropriate points in the narrative. By inviting users to feel the sensation of pain in addition to the narrative’s affective framing of pain, we seek to foster better recognition of early pain signals and raise consciousness about how social context shapes their interpretation.

For later portions of the experience, the user will then be tasked with describing the sensations they feel. This intensity can be controlled through activities such as movement that we measure through the use of additional sensors (e.g. accelerometers) to mimic the biopsychosocial mechanisms through which pain can be attenuated. Future work will include expanding the prototype to encompass more regions of the body.

3.1 Requested BOM

Feature	First Prototype	Second Prototype
Dots per arm	4 dots (vibration + temperature)	—
Chest coverage	2 dots	2 dots on upper chest
Power source(s)	1 shared power source	2 power sources
Cable length	5 ft cables	5 ft cables
3D print files used	Yes	Yes

Table 2: BOM for Pain by Numbers prototypes

Through user testing, DataFeel dot placement experimentation, and participatory co-design with people living with pain, we aimed to build a tool, Pain by Numbers, that expands our collective ability to listen to pain—in ourselves and others. By merging embodied computing, narrative design, and health education, we reframe pain as not a symptom to be silenced, but as a language to be learned.

References

- [1] Hannah Arendt. 1958. *The Human Condition* (2nd ed.). University of Chicago Press, Chicago. 50–51 pages.

- [2] M Catherine Bushnell, Marta Čeko, and Lucina A Low. 2013. Cognitive and emotional control of pain and its disruption in chronic pain. *Nature Reviews Neuroscience* 14, 7 (2013), 502–511.
- [3] James A. Coan, Hilary S. Schaefer, and Richard J. Davidson. 2006. Lending a hand: Social regulation of the neural response to threat. *Psychological Science* 17, 12 (2006), 1032–1039. doi:10.1111/j.1467-9280.2006.01832.x
- [4] Ruth Defrin, Anat Benstein-Sheraizin, Adva Bezalel, Ofira Mantzur, and Lars Arendt-Nielsen. 2008. The spatial characteristics of the painful thermal grill illusion. *Pain* 138, 3 (2008), 577–586.
- [5] Haichen Gao, Shaoyu Cai, Yuhong Wu, and Kening Zhu. 2024. ThermOuch: a wearable thermo-haptic device for inducing pain sensation in virtual reality through thermal grill illusion. In *SIGGRAPH asia 2024 conference papers*. 1–12.
- [6] Philip L. Jackson, Andrew N. Meltzoff, and Jean Decety. 2005. How do we perceive the pain of others? A window into the neural processes involved in empathy. *NeuroImage* 24, 3 (feb 2005), 771–779. doi:10.1016/j.neuroimage.2004.09.006
- [7] Chris J Main. 2016. Pain assessment in context: a state of the science review of the McGill pain questionnaire 40 years on. *Pain* 157, 7 (2016), 1387–1399.
- [8] Sharon L Manne, Paul B Jacobsen, and William H Redd. 1992. Assessment of acute pediatric pain: do child self-report, parent ratings, and nurse ratings measure the same phenomenon? *Pain* 48, 1 (1992), 45–52. doi:10.1016/0304-3959(92)90130-4
- [9] Rilind Shala, Nathalie Roussel, G Lorimer Moseley, Thomas Osinski, and Emilio J Puentedura. 2021. Can we just talk our patients out of pain? Should pain neuroscience education be our only tool? 3 pages.
- [10] Tania Singer, Ben Seymour, John O'Doherty, Holger Kaube, Raymond J. Dolan, and Chris D. Frith. 2004. Empathy for Pain Involves the Affective but not Sensory Components of Pain. *Science* 303, 5661 (feb 2004), 1157–1162. doi:10.1126/science.1093535
- [11] Jarred Younger, Arthur Aron, Stephanie Parke, Nandita Chatterjee, and Sean Mackey. 2010. Viewing pictures of a romantic partner reduces experimental pain: Involvement of neural reward systems. *PLoS ONE* 5, 10 (2010), e13309. doi:10.1371/journal.pone.0013309